

Speak to Awaken EP.1 Diving into Siraya

An Endangered Language Speech Interactive VR Documentary

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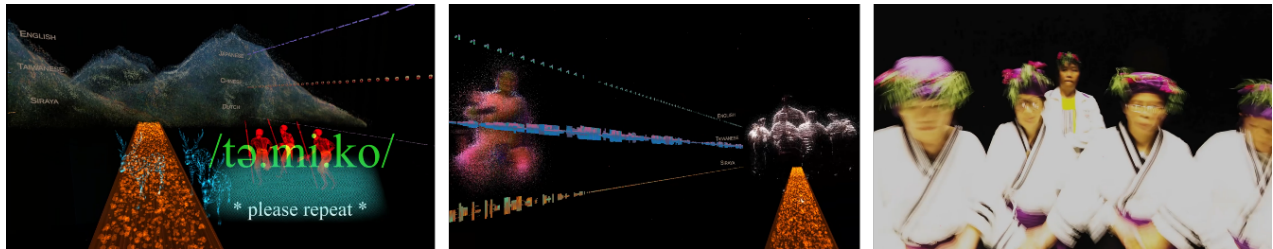


Figure 1: (Left) Word card for speech interaction for Siraya keyword (Middle) Point cloud effect after processing from Volumetric capture of Siraya revitalization character (Right) VR360 video of Night Ceremony

ABSTRACT

Speak to Awaken: Ep.1 Diving into Siraya is an experimental interactive VR documentary that aims to arouse interest in this issue. Experiencers can speak the endangered Siraya language to engage their reconstructed abstract world. They can also absorb facts on the revitalization process through the perspectives of related key persons and graphics. High-resolution VR360 videos, volumetric captures, and together with the voice donation website are adopted, as a new approach for the preservation of endangered language and its culture.

CCS CONCEPTS

• Social and professional topics; • Cultural and historical preservation; • Human-centered computing; • Visualization; • Empirical studies in visualization;

KEYWORDS

Speech interaction, Virtual reality, VR360, Endangered language

ACM Reference Format:

Carol Tsai-Jung Han and Kuan-Yuan Lai. 2021. Speak to Awaken EP.1 Diving into Siraya: An Endangered Language Speech Interactive VR Documentary. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH '21 Immersive Pavilion)*, August 09–13, 2021, Virtual Event, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3450615.3464537>

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SIGGRAPH '21 Immersive Pavilion, August 09–13, 2021, Virtual Event, USA

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ACM ISBN 978-1-4503-8368-4/21/08.

<https://doi.org/10.1145/3450615.3464537>

1 THE BACKGROUND

The Sirayan language is spoken by the Taiwanese plains indigenous people of the Siraya Tribe. During 1629-1662 AD, Dutch missionaries Romanized Siraya language in an attempt to Christianize the population, therefore bring about far-reaching structural change in their communities. In the 19th century, due to the immigration of the Hans from China, the culture of the Siraya people had gradually been Sinicized. Their the language and traditions had been lost. In recent years, there has been a movement of the rectification of aboriginal names and terms. After two decades of efforts of revitalization by the Tainan City Government Ethnic Affairs Commission, Sirayan language has finally been in use again. However, it has still been listed as extinct by UNESCO until now. Even though, it is a rare case in comparison to other extinct languages worldwide. It becomes, therefore, an inspiring and intuitive example for this movement.

50% of the world's languages are endangered, every two weeks a language falls silent. Language preserves the cultures, the historical memories, so as the ethnic identities, as well as the unique worldviews upon their social and environmental context. These worldviews act as wisdoms or remedies for the difficulties during that ages. The importance of keeping linguistic diversity is just comparable to the necessity of preserving bio-diversity. An underlying dilemma is which is the representative story for this topic. A series of works that represent the effort of people with different language status under different conditions, are probably more capable to depict the complexity and diversity of this endangerment issue, and manifest the value of their individual intellectual heritages.

We thus have the Speak to Awaken series. Speak to Awaken is a series project of speech interactive virtual reality experimental film on endangered language. There is a website of the same name as the common portal for descriptions on the series and on each episode. Visitors are allowed to donate their voice on keywords of

each episode to help build the speech recognition machine learning model for future usages.

2 THE ARTWORK

Diving into Siraya is an abstract journey on the Sirayan aboriginal language in Taiwan, that has been undergoing attempts at revitalization.

Following the guidance of the keyword floating above the animated cloud, pronunciations with variant accents and tones are spreading in the space, the audience can proceed along the road as encouragement if willing to speak. The road symbolized the embodiment of Heidegger's *On the Way To Language*. Once succeeded, both the corresponding visual and audio effects are unlocked. In some scenes, the delicate cultural behavior is triggered as a representative of Siraya's unique worldviews. This process mimicking the mother tongue childhood language learning experience in a more natural way than classroom annotations. Through the relevant feedback effects, the player's curiosity is aroused for estimating the real meaning of the keyword. They are invited to carve the world as a member of Sirayan.

The Sirayan had been transformed by several populations during different periods. Events relevant to that period are spoken out in corresponding language, and visualize with fluctuation of representative soundwaves aside as the testimony of the time.

Siraya language has been lost for more than two hundred years, and has been reconstructed through literature and research. Virtual 3D reconstructions thus adopt more frequently in the former part of the work, while VR 360 and traditional shootings are more common to represent the reality in the latter part. The spiritual part of the interviews is processed as point cloud effects, which may disperse in the space in the form of dynamic points and bubbles. In audio effects, particle synthesis techniques are used to design the sound effects and tone colors in musical passages.

Traditional territories and cultural events are the connections between history and present. The audience can pass through them after the interaction or narratives. In addition to the language, the precious Night Sacrifice Ceremony, together with the only surviving, incomprehensible Qianqu songs are revitalized. Converging from tiny efforts to concrete reality, the efforts of Sirayan eventually developed into to a vivid scene.

3 THE TECHNOLOGIES

VR itself is a resource-heavy media. In Diving into Siraya, the situation becomes even worse due to the integration of complex media sources and requirements. There are three limitations in this project: Speech recognition, point cloud effect from 3D and volumetric capture [Elayat et al., 2017], and high-resolution VR360. Efficient usage of CPU and GPU across scene is the key to a smooth experience in this work.

3.1 Speech recognition

To achieve the challenge of speech recognition with sufficient stability, sensitivity, accuracy, under the condition of high system resource consumption of VR, we tried several options like self-build model, optimize pre-trained model, both online and offline solutions. Further constraints of endangered language accessibilities

including the audio quality, sufficient data amount, and the cost of training individual keywords, etc. Eventually, we take advantage on the language package in win10 and HTC Vive Pro 2 built-in directional noise-reduction microphone as our scheme. The language packages take very little system resource and realize streaming recognition with low latency sufficient for smooth speech interaction. Keywords can easily be added or removed in response to plot developments. Furthermore, the regional-optimized packages provide the basis for accents, tones, pronunciations variations from different regions or populations, which serve not only the recognition precision but also fit the trend of inclusion in the museum exhibition.

3.2 Point cloud effect

The Point cloud effect is the symbolization of the virtual world and spirit of the Siraya people. This effect should be approached according to the original data type. The main challenge on this part is the balance between artistic aesthetic and computation power. For the 3D models, only a specific shader of the universal render pipeline is used for batch computing. Tiny adjustments such as texture color and tilt are allowed to keep the visual consistency while enriching with diversity. Volumetric captures with depth maps and volumetric captures with point clouds are processed separately in advance, for the sake that these two systems have their own render pipeline and processing workflow. Images from each system then undergo the same visual effect treatments to balance the visual inconsistency before being sent into the main project for integration.

3.3 High-resolution VR 360 video

We use up to 8K of the VR360 for the shooting. Image quality resulting from the optical quality of the original camera and shooting condition, stitching quality, transfer quality of the headset cables are all issues on this part. However, the main limitation is how to play the high-resolution VR video smoothly with artistic and visual quality consideration. Image effect and rectification treatments follow with proper setting of H.265 transcode setting solve this issue.

ACKNOWLEDGMENTS

We appreciate Meng-Jen TUNG, Pierre Magistry, Fungs design, and Yi-Cheng Zoe LIN as the production, technology and creative team of this project; NowHere media as our international partner. This work was supported by the 2020 Kaohsiung Film Archive "VR FILM LAB" and CNEX Chinese Documentary Forum (CCDF).

REFERENCES

Yasmin Elayat, Elie Zananiri, Mei-Ling Wong, Alexander Porter, James George. Zero Days VR. SIGGRAPH '17: ACM SIGGRAPH 2017 Computer Animation Festival. July 2017 Pages 37 <https://doi.org/10.1145/3078280.3105235>